Bounds for $\mathrm{Tr}(M(A \otimes A))$ via the TT-sketch chain construction: summary of progress and open problems

Abstract

This note collects what has been established analytically and what remains open about the bilinear form $\mathrm{Tr}(M(A \otimes A))$, where M is the second-moment matrix of the tensor-train (TT) sketch and A is a $d^N \times d^N$ matrix. Section 2 states the verified results: an exact closed-form expression valid for any matrix-product-operator (MPO) representation of A, and a clean variance bound for diagonal PSD A. Section 3 states what is missing: extending the diagonal bound to general PSD A in a way that beats Avron's 3^N benchmark without further structural assumptions on A. Numerical evidence strongly supports the conjecture that the bound holds in the general PSD case, but no analytical proof is in hand.

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1 Setup and notation

Fix integers $m \geq 1$, $N \geq 3$, $d \geq 2$, and let $D = d^N$. The TT sketch produces a matrix $S \in \mathbb{R}^{m \times D}$ with $\mathbb{E}[S^\top S] = I_D$. The second-moment matrix is

$$M\big[(i^a, i^c), (i^b, i^d)\big] \;=\; \mathbb{E}\big[(S^\top S)_{i^a i^b}\,(S^\top S)_{i^c i^d}\big] \quad \in \mathbb{R}^{D^2 \times D^2},$$

with rows indexed by $(i^a, i^c) \in [d]^N \times [d]^N$ and columns by (i^b, i^d).

For any (deterministic) matrix $A \in \mathbb{R}^{D \times D}$, set $X(A) = \langle S^\top S,\, A\rangle = \mathrm{tr}(SAS^\top)$. Then $\mathbb{E}[X(A)] = \mathrm{tr}(A)$ and the second moment is

$$\mathbb{E}[X(A)^2] \;=\; \mathrm{Tr}\big(M\,(A \otimes A)\big), \qquad \mathrm{Var}\big(X(A)\big) \;=\; \mathrm{Tr}\big(M(A \otimes A)\big) - \mathrm{tr}(A)^2.$$

The goal is to bound $\mathrm{Tr}(M(A \otimes A))$ as a function of A, m, N, d.

The four *replica patterns* $\boldsymbol{P} = \{4K, A, U_1, U_2\}$ are the four even-block partitions of $\{a, b, c, d\}$:

$$\begin{aligned} 4K &: \{a = b = c = d\}, & A &: \{a = b\} \sqcup \{c = d\}, \\ U_1 &: \{a = c\} \sqcup \{b = d\}, & U_2 &: \{a = d\} \sqcup \{b = c\}. \end{aligned}$$

M admits a chain decomposition over pattern sequences $\boldsymbol{P}^N$, with per-site 4×4 pattern transfer matrices W^P and boundary vectors $u_{\mathrm{init}}^{(P_1,P_2)}, u_{\mathrm{end}}^{P_N} \in \mathbb{R}^4$, all explicit polynomials in $(m, m_1, m_2, \sigma_2, \sigma_4)$ (formulas (19)–(21) of the thesis). The variance and trace lemmas of the thesis correspond to specific aggregations of these blocks.

2 Results established

2.1 Closed-form $\mathrm{Tr}(M(A \otimes A))$ for MPO A

Theorem 1 (Closed form, verified to machine precision)**.** *Suppose A is given as an MPO with cores $A^{(l)} \in \mathbb{R}^{r_{l-1} \times d \times d \times r_l}$, $l = 1, \ldots, N$, with $r_0 = r_N = 1$ and bond dimensions $r_1, \ldots, r_{N-1}$. For each site l and each pattern $P \in \boldsymbol{P}$, define the $r_{l-1}^2 \times r_l^2$ matrix*

$$\beta_P^{(l)}\big[(\alpha, \gamma), (\alpha', \gamma')\big] \;=\; \sum_{(a,b,c,d) \in P} A_{ab}^{(l)}[\alpha, \alpha']\, A_{cd}^{(l)}[\gamma, \gamma']. \tag{1}$$

With $\widetilde{W}^{(l)} = \sum_P W^P \otimes (\beta_P^{(l)})^\top$ and the analogous boundary vectors,

$$\mathrm{Tr}\big(M(A \otimes A)\big) \;=\; \widetilde{u}_{\mathrm{end}}^\top\, \widetilde{W}^{(N-1)} \cdots \widetilde{W}^{(3)}\, \widetilde{u}_{\mathrm{init}}.$$

Remark 2. *The proof and explicit definitions of $\widetilde{u}_{\mathrm{init}}$, $\widetilde{u}_{\mathrm{end}}$, $\widetilde{W}$ are given in detail in the companion note* "A closed-form expression for $\mathrm{Tr}(M(A \otimes A))$ when A is given as a Matrix Product Operator"*. A reference implementation (`trace_MB_mpo.py`) verifies the formula against* $\mathrm{vec}(A)^\top M\ \mathrm{vec}(A)$ *for the materialized $D^2 \times D^2$ matrix M on six configurations, with relative error $\sim 10^{-16}$.*

Cost. With $r := \max_l r_l$, the formula evaluates in $O(Nr^4)$ floating-point operations. No assumption on A beyond MPO representability (which holds for every matrix, with r growing as needed).

2.2 Diagonal PSD A: Avron-comparable bound

Theorem 3 (Diagonal case, clean bound)**.** *Let $A \in \mathbb{R}^{D\times D}$ be diagonal in the product basis, $A = \mathrm{diag}(\lambda_1, \ldots, \lambda_D)$ with $\lambda_i \geq 0$. Then for site-uniform sketch parameters,*

$$\mathrm{Tr}\big(M(A \otimes A)\big) \;=\; \widetilde{u}_e^{\top}\, \widetilde{W}_{\mathrm{diag}}^{N-3}\, \widetilde{u}_i,$$

where $\widetilde{W}_{\mathrm{diag}}$ is a 4×4 matrix with only the $\{4K, A\}$ pattern columns active (the U_1, U_2 contributions vanish), and a spectral analysis of $\widetilde{W}_{\mathrm{diag}}$ yields

$$\boxed{\mathrm{Tr}\big(M(A \otimes A)\big) \;\leq\; \big(1 + c(d)/m\big)^N\, \mathrm{tr}(A)^2} \tag{2}$$

with $c(d)$ the same constant extracted numerically for $A = I$: $c(2) \approx 1.05$, $c(3) \approx 0.82$, $c(4) \approx 0.61$, $c(5) \approx 0.49$.

Sketch of proof. For diagonal A, the 4-tuple constraint $(i_l^a, i_l^b, i_l^c, i_l^d) \in P_l$ combined with the diagonality ($A_{i_l^a i_l^b}\, A_{i_l^c i_l^d} = \delta_{i_l^a, i_l^b} \delta_{i_l^c, i_l^d}\, \lambda_{i_l^a} \lambda_{i_l^c}$) forces $i_l^b = i_l^a$ and $i_l^d = i_l^c$ at every site. The only patterns consistent with this are $4K$ (when $i_l^a = i_l^c$) and A (when $i_l^a \neq i_l^c$). Plugging into the chain identity (1)–Theorem 1 reduces $\beta_P^{(l)}$ to a scalar at each site: $\beta_{4K}^{(l)} = \sum_a \lambda_{a,l}^2$, $\beta_A^{(l)} = (\sum_a \lambda_{a,l})^2 - \beta_{4K}^{(l)}$, $\beta_{U_1}^{(l)} = \beta_{U_2}^{(l)} = 0$, where $\lambda_{a,l}$ is the contribution at site l to λ_i.

Denoting the leading eigenvector of $\widetilde{W}_{\mathrm{diag}}$ (the "mean" direction) by v_*, the chain identity $\widetilde{u}_e^{\top} \widetilde{W}^{N-3} \widetilde{u}_i$ telescopes: the leading eigenvalue λ_* contributes $\lambda_*^{N-3}\, \langle \widetilde{u}_e, v_* \rangle \langle v_*, \widetilde{u}_i \rangle = \mathrm{tr}(A)^2$ (this is the unbiasedness identity), and sub-leading eigenvalues $\lambda_2 < \lambda_*$ contribute $O((\lambda_2/\lambda_*)^{N-3})\, \mathrm{tr}(A)^2$. Setting $1 + c(d)/m = \lambda_2/\lambda_*$ gives the bound. □ □

Remark 4. *The bound (2) matches the empirically extracted form for $A = I$ exactly, with the same constant $c(d)$. It is the cleanest analogue of Avron's $\mathrm{tr}(A)^2$ bound for the TT sketch.*

2.3 Comparison with Avron

For Kronecker-Hutchinson (KH) sketches with ℓ samples and PSD A, Avron's bound [1] gives

$$\mathrm{Var}\big(\tfrac{1}{\ell}\langle x^{\top} A x \rangle\big) \;\leq\; \frac{(1 + 2/d)^N - 1}{\ell}\, \mathrm{tr}(A)^2 \;\leq\; \frac{3^N}{\ell}\, \mathrm{tr}(A)^2.$$

This requires $\ell = \Omega(3^N/\varepsilon^2)$ samples for ε relative error.

For the TT sketch with diagonal PSD A, Theorem 3 gives

$$\mathrm{Var}\big(\langle S^{\top} S, A \rangle\big) \;\leq\; \big((1 + c(d)/m)^N - 1\big)\, \mathrm{tr}(A)^2 \;\sim\; \frac{N\, c(d)}{m}\, \mathrm{tr}(A)^2 \quad (m \gg N).$$

This requires $m = \Omega(N\, c(d)/\varepsilon^2)$ for ε relative error — *linear* in N, no exponential blow-up.

The TT sketch is therefore exponentially more efficient than KH in the diagonal-A regime: $m = O(N)$ vs $\ell = O(3^N)$ for the same accuracy.

3 What remains open

3.1 General PSD A: numerical evidence vs analytical proof

For general (non-diagonal) PSD A, the off-diagonal MPO slices $A_{ab}^{(l)}$ ($a \neq b$) activate the U_1 and U_2 patterns in the chain. Empirically, the bound (2) continues to hold across all PSD A tested:

- For random PSD A with bond dimension up to $r = 3$ and $(m, N, d) \in \{(4, 3, 2), (4, 4, 2), (4, 5, 2), (6, 4, 3)\}$, direct evaluation of $\mathrm{Tr}(M(A \otimes A))$ via Theorem 1 gives values consistent with $(1+c(d)/m)^N \,\mathrm{tr}(A)^2$.
- Numerical exploration of the worst-direction Rayleigh quotient $\mathrm{Tr}(M(A \otimes A))/\,\mathrm{tr}(A)^2$ over PSD A with $\mathrm{tr}(A) = 1$ does not exceed $(1 + c(d)/m)^N$ in any tested configuration.

Conjecture 5 (Open). *For every PSD $A \in \mathbb{R}^{D \times D}$,*

$$\mathrm{Tr}\big(M(A \otimes A)\big) \;\le\; (1 + c(d)/m)^N \;\mathrm{tr}(A)^2,$$

with $c(d)$ the same constant as in the diagonal case.

3.2 Why the diagonal proof does not directly extend

The diagonal case works because the index constraints $i_l^b = i_l^a$ and $i_l^d = i_l^c$ at every site force only $4K$ and A patterns, and the per-site sums $\beta_{4K}^{(l)}$, $\beta_A^{(l)}$ telescope into $\sum_i \lambda_i^2$ and $\mathrm{tr}(A)^2 - \sum_i \lambda_i^2$ respectively, both of which are bounded by $\mathrm{tr}(A)^2$.

For non-diagonal PSD A, the off-diagonal entries A_{ij} $(i \neq j)$ enter $\beta_{U_1}^{(l)}$ and $\beta_{U_2}^{(l)}$ as $\sum_{a \neq b} A_{ab}^{(l)} \otimes A_{ab}^{(l)}$ and similar. Telescoping these into a global A-norm requires control of the bond-index structure across sites, which the off-diagonal MPO entries do not provide cleanly. Specifically:

- The natural per-site bound $\|\beta_{U_1}^{(l)}\|_F \le \|A^{(l)}\|_F^2$ telescopes into $\|A\|_F^{2N}$, which is much larger than $\mathrm{tr}(A)^2$ when N is large and the off-diagonals are dense.
- Tighter per-site bounds in terms of the operator norm of $\beta_{U_1}^{(l)}$ require analysis of the joint MPO spectrum, which reduces to a non-trivial spectral problem on a r^2-dimensional bond space.
- The "right" object to bound is the worst-direction Rayleigh quotient of M on the cone of PSD A with $\mathrm{tr}(A) = 1$, which empirically is achieved at $A = I$ (the maximally symmetric direction) but proving this requires representation-theoretic arguments specific to the TT sketch's symmetry group.

3.3 The A-as-loose-bound issue

A complementary partial result, valid for any MPO A with bounded core Frobenius norm $\|A_{ab}^{(l)}\|_F \le K$, is

$$\big|\mathrm{Tr}(M(A \otimes A))\big| \;\le\; (dK^2)^N \,(1 + c(d)/m)^N.$$

This gives the friendly $(1 + c/m)^N$ tensor-train factor but the prefactor $(dK^2)^N$ is loose: it bounds $\mathrm{tr}(A)^2$ from above generally only by an exponential factor in N. The bound is therefore weaker than Avron's $3^N \,\mathrm{tr}(A)^2$ for large d, and is not the correct shape for the desired comparison.

3.4 What would close the gap

Three potential routes to extending Theorem 3 to general PSD A:

Route 1: representation theory. Show that for any PSD A with $\mathrm{tr}(A) = 1$, the Rayleigh quotient $\mathrm{vec}(A)^\top M \,\mathrm{vec}(A)$ is maximized at $A = I/D$ (the symmetric direction). Combined with the diagonal-case bound at $A = I$, this would give the conjecture immediately.

The TT sketch is invariant under permutations of the d indices at each site and under interchange of replicas $a \leftrightarrow c$, $b \leftrightarrow d$. This forces M's top eigendirection on the PSD cone to lie in the symmetric subspace, and $A = I$ is the unique fixed point of the full symmetry group. A representation-theoretic argument should make this precise but has not been carried out.

Route 2: structural assumption on A. Restrict to PSD A that is diagonal in some product basis: $A = U^{\otimes N} D U^{\otimes N\top}$ for D diagonal and U a $d \times d$ unitary. The sketch is U-equivariant if U acts trivially on the sketch's symmetry group, in which case Theorem 3 applies in the rotated basis. This is a non-trivial constraint but is satisfied by some natural A in applications.

Route 3: direct chain analysis. Prove the conjecture by a direct spectral analysis of the 4×4 aggregated chain matrix $\widetilde{W}$ for the worst PSD direction. The chain has only four patterns and the boundary-projection technique that works for the diagonal case may extend with more careful bookkeeping. This is the most laborious but most direct route.

4 Application-specific note: A as a reduced density matrix

The user's stated application uses A as a reduced density matrix (RDM) of a quantum many-body state. RDMs have several structural properties relevant here:

- PSD with $\mathrm{tr}(A) = 1$ (by definition).
- In an entanglement-eigenbasis, A is diagonal — so for applications that compute traces or expectation values in this basis, Theorem 3 applies directly with the bound (2).
- For RDMs of low-entanglement states (matrix-product states), A has low MPO bond dimension, so even the closed-form Theorem 1 is computationally cheap.

If the application can be formulated in the diagonal basis of the RDM, **Theorem 3 suffices and gives an exponential improvement over Avron**. If the application requires evaluating $\mathrm{Tr}(M(A \otimes A))$ in a fixed (non-diagonal) basis, Conjecture 5 is needed — the empirical evidence supports it but the proof is open.

5 Summary

Established.

- Exact closed-form $\mathrm{Tr}(M(A \otimes A))$ via $4r^2$-dimensional chain for any MPO A (Theorem 1, verified to machine precision).
- Avron-comparable bound $(1 + c(d)/m)^N \,\mathrm{tr}(A)^2$ for diagonal PSD A (Theorem 3). This is exponentially better than KH's 3^N.

Conjectured but unproven.

- The same bound for arbitrary PSD A (Conjecture 5). Numerical evidence supports it; an analytical proof requires either representation theory, structural assumptions, or a more careful chain analysis.

Recommendation. For the RDM application: state the result for diagonal A (or equivalently, in the eigenbasis of A). This covers the most natural use cases for RDMs and gives the strongest clean bound. Note that empirically the bound holds for general PSD A and leave the latter as an open problem.

A Approaches that were tried and rejected

This appendix records analytical paths that were considered and ruled out, to save time in continuation work.

A.1 Row-sum bound (the original thesis approach)

The thesis row-sum lemma states

$$\mathrm{Tr}(MB) \;\le\; R \cdot \mathrm{Tr}(B), \qquad R := \max_I \sum_J M_{IJ}.$$

For $B = A \otimes A$ with A PSD, $\mathrm{Tr}(B) = \mathrm{tr}(A)^2$, so $\mathrm{Var}(\langle S^\top S, A\rangle) \le (R-1)\,\mathrm{tr}(A)^2$.

Rejected because: numerical evidence shows $R - 1 \sim \alpha(d)^N/m$ with $\alpha(d) \approx d$. The bound is therefore exponential in N and loses to Avron's 3^N. The diagnostic plot at $m = N$ confirms: $\log(1 + (R-1))$ grows linearly with N (slope $\log d$), whereas $\log(1 + \mathrm{Tr}(M)/d^{2N} - 1)$ asymptotes to a constant (the friendly $(1 + a/m)^N$ form). The row-sum step is loose by a factor of roughly d^N.

A.2 Operator norm $\|M\|_{\mathrm{op}}$

Bound: $\mathrm{Tr}(MB) \le \|M\|_{\mathrm{op}}\,\mathrm{Tr}(B) = \|M\|_{\mathrm{op}}\,\mathrm{tr}(A)^2$ for PSD M, B.

Rejected because: computing $\|M\|_{\mathrm{op}}$ requires diagonalizing M, which is a $d^{2N} \times d^{2N}$ matrix. Even via power iteration on the chain, the matrix-vector product Mv for a general $v \in \mathbb{R}^{d^{2N}}$ does not preserve a tractable structure, so power iteration would still require materializing v in full. Feasible only for very small (N, d).

The same applies to bounds via $\sup_u V_u$ where $V_u = (u \otimes u)^\top \mathcal{C}(u \otimes u)$ for unit u: this quantity is operator-norm-grade hard to compute for general u, and restricting to tensor-structured u gives only a lower bound on the quantity needed.

A.3 Frobenius bound via $\sqrt{\mathrm{Tr}(M^2)}$

Bound: $\mathrm{Tr}(MB) \le \|M\|_F\,\|B\|_F = \sqrt{\mathrm{Tr}(M^2)}\,\|A\|_F^2$.

Rejected because: (i) $\mathrm{Tr}(M^2)$ requires an 8-replica chain construction (pattern space of size ~ 15 instead of 4), which is analytically derivable but is non-trivial new work, and (ii) the bound scales with $\|A\|_F^2$ rather than $\mathrm{tr}(A)^2$, which is the wrong shape for an Avron-comparable result and loses up to a factor of $\mathrm{rank}(A)$.

A.4 Trace bound: $\mathrm{Tr}(MB) \le \mathrm{Tr}(M)\|A\|_{\mathrm{op}}^2$

Holds for PSD M, B via Schatten 1-∞ duality.

Rejected because: the prefactor is $\mathrm{Tr}(M)$, which scales as $d^{2N}(1 + a(d)/m)^N$. Even though the $(1 + a(d)/m)^N$ part is correct, the d^{2N} prefactor combined with $\|A\|_{\mathrm{op}}^2$ gives a bound much weaker than the row-sum bound, hence weaker than Avron.

A.5 Cauchy–Schwarz on covariances + eigendecomposition of A

For symmetric $A = \sum_i \lambda_i u_i u_i^\top$,

$$\mathrm{Var}(\langle S^\top S, A\rangle) \le \Big(\sum_i |\lambda_i|\Big)^2 \sup_{\|u\|=1} V_u.$$

Rejected because: $\sup_{\|u\|=1} V_u$ is operator-norm-grade hard. The bound effectively reduces to the operator norm and is no better than the operator-norm route.

A.6 Per-site bound by core Frobenius norm

For MPO A with $\|A^{(l)}_{ab}\|_F \le K$ for all a, b, l:

$$|\operatorname{Tr}(M(A \otimes A))| \le (dK^2)^N (1 + c/m)^N.$$

Rejected because: the $(dK^2)^N$ prefactor does not telescope into $\mathrm{tr}(A)^2$ for general MPO A. It bounds $\mathrm{tr}(A)^2$ only by an exponential factor in N. Slightly worse than Avron for large d. The $(1 + c/m)^N$ factor is correct, but the prefactor is too loose to demonstrate the win against $\mathrm{tr}(A)^2$.

A.7 Common error: confusing absolute and relative variance

Several diagnostic plots and bound attempts were initially confused because $\mathrm{Tr}(M)$ has a natural scale of $d^{2N} = \mathrm{tr}(I)^2$, while $R - 1$ is already in $\mathrm{tr}(A)^2$ units. The discriminating plot is $\log(1 + \mathrm{Tr}(M)/d^{2N} - 1)$ vs $\log(1 + (R - 1))$, both at $m = N$; the first asymptotes (friendly form, good), the second grows linearly (bad form, bad).

Plotting $\log(1 + \mathrm{Tr}(M))$ alone gives a straight line growing with slope $2 \log d$ — this is just the natural scale and does *not* indicate the bound is bad. Always normalize by $\mathrm{tr}(A)^2$ before checking the form of the bound.

A.8 Bug in the original row-sum reference code

The thesis's original `compute_max_row_sum` reference code had an off-by-one factor of m in entry $WU_1[A, A]$. The corrected entry is

$$W^{U_1}[A, A] \;=\; m^4(m-1)\sigma_2^6 \quad \big(\text{not } m^3(m-1)\sigma_2^6\big).$$

This was diagnosed by Monte Carlo verification at $(m, N, d) = (4, 6, 2)$ rademacher: the corrected formula matches MC to $8{\times}10^{-4}$ relative error; the original was off by 2.6% and grew with N. Verify any thesis row-sum code has this fix before trusting numerical results.

A.9 Approaches not yet attempted

- Representation-theoretic argument for $A = I$ being the worst PSD direction in $\mathrm{tr}(A)^2$ normalization. The TT sketch's symmetry group (permutations $\times$ replica interchange) has $\mathrm{vec}(I)$ as its unique fixed-point on the PSD cone modulo scale; making this rigorous via Schur–Weyl-type decomposition is the most promising open direction.
- Direct 4×4 chain analysis showing $\lambda_{\max}(\widetilde{W}) = 1 + a(d)/m$ for the worst PSD-direction boundary. A spectral analysis of the boundary projection onto the top eigenvector is the technical bottleneck.

- Lifting one-vector JL bounds (Ahle et al., Rakhshan–Rabusseau) from $\|x\|^4$ to $\mathrm{tr}(A)^2$ for matrices: the known proofs use decoupling arguments specific to single-vector statements that don't obviously extend.

References

[1] H. Avron and R. Meyer, *Hutchinson trace estimation for matrix-vector products with Kronecker structure*, arXiv:2309.04952, 2023.